

Question 21. From the EMP table show the minimum, maximum and average basic for each department (show dept. Code).

Query :

```
SELECT
    DEPT_CODE,
    MIN(BASIC) AS MIN_BASIC,
    MAX(BASIC) AS MAX_BASIC,
    AVG(BASIC) AS AVG_BASIC
FROM EMPLOYEE
GROUP BY DEPT_CODE;
```

```
mysql> SELECT
```

```
->     DEPT_CODE,
->     MIN(BASIC) AS MIN_BASIC,
->     MAX(BASIC) AS MAX_BASIC,
->     AVG(BASIC) AS AVG_BASIC
-> FROM EMPLOYEE
-> GROUP BY DEPT_CODE;
```

DEPT_CODE	MIN_BASIC	MAX_BASIC	AVG_BASIC
DP001	19106.00	89977.00	55118.742574
DP002	19451.00	89732.00	54488.371859
DP003	18230.00	89438.00	53906.551724
DP004	18251.00	89935.00	54143.875598
DP005	18000.00	89914.00	55452.692708

```
5 rows in set (0.06 sec)
```

Question 22. Find the number of female employees in each department (show dept. Code).

Query :

```
SELECT
    DEPT_CODE,
    COUNT(*) AS FEMALE_EMPLOYEES
FROM EMPLOYEE
WHERE SEX='F'
GROUP BY DEPT_CODE;
```

```
mysql> SELECT
```

```
->     DEPT_CODE,
->     COUNT(*) AS FEMALE_EMPLOYEES
-> FROM EMPLOYEE
-> WHERE SEX='F'
-> GROUP BY DEPT_CODE;
```

DEPT_CODE	FEMALE_EMPLOYEES
DP001	94
DP002	97
DP003	98
DP004	119
DP005	96

```
5 rows in set (0.00 sec)
```

Question 23: Find the city wise no. of employees for each department (show dept. Code).

Query :

```
SELECT
    DEPT_CODE,
    CITY,
    COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
GROUP BY DEPT_CODE, CITY;
```

```
mysql> SELECT
->     DEPT_CODE,
->     CITY,
->     COUNT(*) AS TOTAL_EMPLOYEES
-> FROM EMPLOYEE
-> GROUP BY DEPT_CODE, CITY;
```

DEPT_CODE	CITY	TOTAL_EMPLOYEES
DP002	Guwahati	3
DP001	Murshidabad	4
DP001	KOLKATA	11
DP005	Kharagpur	6
DP002	Kolkata	3
DP002	Agra	2
DP001	Chandigarh	9
DP005	Patna	3
DP004	Faridabad	6
DP002	Bardhaman	7
DP004	Gurugram	6
DP002	Pune	7
DP004	Chandigarh	4
DP005	Madurai	5
DP004	Mumbai	4
DP002	Jalpaiguri	1

Question 24 : Show the designation wise no of employees who have joined in the year 2000 in each department. The listing should appear in the ascending order of no. of employees.

Query :

```
SELECT
    E.DEPT_CODE,
    D.DESIG_DESC,
    COUNT(*) AS EMP_COUNT
FROM EMPLOYEE E
JOIN DESIGNATION D
    ON E.DESIG_CODE=D.DESIG_CODE
WHERE YEAR(E.JN_DT)=2000
GROUP BY E.DEPT_CODE,D.DESIG_DESC
ORDER BY EMP_COUNT ASC;
```

```
mysql> SELECT
->     E.DEPT_CODE,
->     D.DESIG_DESC,
->     COUNT(*) AS EMP_COUNT
-> FROM EMPLOYEE E
-> JOIN DESIGNATION D
->     ON E.DESIG_CODE=D.DESIG_CODE
-> WHERE YEAR(E.JN_DT)=2000
-> GROUP BY E.DEPT_CODE,D.DESIG_DESC
-> ORDER BY EMP_COUNT ASC;
Empty set (0.02 sec)
```

Question 25: Find the department code wise total basic of male employees only for the departments for which such total is more than 50,000 and the listing should appear in the descending order of total basic.

Query :

```
SELECT
    DEPT_CODE,
    SUM(BASIC) AS TOTAL_BASIC
FROM EMPLOYEE
WHERE SEX='M'
GROUP BY DEPT_CODE
HAVING SUM(BASIC) > 50000
ORDER BY TOTAL_BASIC DESC;
```

```
mysql> SELECT
->     DEPT_CODE,
->     SUM(BASIC) AS TOTAL_BASIC
-> FROM EMPLOYEE
-> WHERE SEX='M'
-> GROUP BY DEPT_CODE
-> HAVING SUM(BASIC) > 50000
-> ORDER BY TOTAL_BASIC DESC;
```

DEPT_CODE	TOTAL_BASIC
DP001	6224335.00
DP003	5635163.00
DP002	5564516.00
DP004	5089610.00
DP005	4952076.00

5 rows in set (0.00 sec)

Question 26: Show the employee name, Designation description and basic for all employees.

Query :

```
SELECT
    E.EMP_NAME,
    D.DESIG_DESC,
    E.BASIC
FROM EMPLOYEE E
JOIN DESIGNATION D
    ON E.DESIG_CODE=D.DESIG_CODE;
```

```
mysql> SELECT
->     E.EMP_NAME,
->     D.DESIG_DESC,
->     E.BASIC
-> FROM EMPLOYEE E
-> JOIN DESIGNATION D
->     ON E.DESIG_CODE=D.DESIG_CODE;
```

EMP_NAME	DESIG_DESC	BASIC
Rahul Das	Manager	60000.00
Deep Khanna	Manager	61575.00
Biswajit Yadav	Manager	53196.00
Rahul Yadav	Manager	33396.00
Rakesh Menon	Manager	34600.00
Sudip Mitra	Manager	38001.00
Ankita Sharma	Manager	22993.00
Avijit Ganguly	Manager	24922.00
Poulomi Sarkar	Manager	82879.00
Santanu Pramanik	Manager	31511.00
Prasenjit Barman	Manager	54864.00
Priya Mehta	Manager	39760.00

Question 27 : Show the employee name, Designation description, Department Name & Basic for all employees.

Query :

```
SELECT
    E.EMP_NAME,
    DG.DESIG_DESC,
    DP.DEPT_NAME,
    E.BASIC
FROM EMPLOYEE E
JOIN DESIGNATION DG
    ON E.DESIG_CODE=DG.DESIG_CODE
JOIN DEPARTMENT DP
    ON E.DEPT_CODE=DP.DEPT_CODE;
```

```
mysql> SELECT
->     E.EMP_NAME,
->     DG.DESIG_DESC,
->     DP.DEPT_NAME,
->     E.BASIC
-> FROM EMPLOYEE E
-> JOIN DESIGNATION DG
->     ON E.DESIG_CODE=DG.DESIG_CODE
-> JOIN DEPARTMENT DP
->     ON E.DEPT_CODE=DP.DEPT_CODE;
```

EMP_NAME	DESIG_DESC	DEPT_NAME	BASIC
Gourav Poddar	Executive	Personnel	28212.00
Rahul Das	Manager	Personnel	60000.00
Akash Nandi	Executive	Personnel	65052.00
Madhumita Naidu	Executive	Personnel	54267.00
Rakesh Singh	Clerk	Personnel	61017.00
Bikash Gupta	Officer	Personnel	30660.00
Tania Debnath	Clerk	Personnel	20725.00
Debjani Poddar	Officer	Personnel	64025.00
Rakesh Menon	Manager	Personnel	34600.00
Rahul Nair	Executive	Personnel	50785.00

Question 28 : Find the department Codes in which no employee works.

Query :

```
SELECT DEPT_CODE
FROM DEPARTMENT
WHERE DEPT_CODE NOT IN
(
    SELECT DISTINCT DEPT_CODE
    FROM EMPLOYEE
    WHERE DEPT_CODE IS NOT NULL
);
```

```
mysql> SELECT DEPT_CODE
-> FROM DEPARTMENT
-> WHERE DEPT_CODE NOT IN
-> (
->     SELECT DISTINCT DEPT_CODE
->     FROM EMPLOYEE
->     WHERE DEPT_CODE IS NOT NULL
-> );
Empty set (0.00 sec)
```

Question 29 : Find the department names where at least one employee works.

Query :

```
SELECT DISTINCT D.DEPT_NAME
FROM DEPARTMENT D JOIN EMPLOYEE E ON D.DEPT_CODE=E.DEPT_CODE;
```

```
mysql> SELECT DISTINCT D.DEPT_NAME
-> FROM DEPARTMENT D
-> JOIN EMPLOYEE E
-> ON D.DEPT_CODE=E.DEPT_CODE;
```

```
+-----+
| DEPT_NAME |
+-----+
| Personnel |
| Production|
| Purchase  |
| Finance   |
| Research  |
+-----+
```

5 rows in set (0.00 sec)

Question 30: Find the department names where at least 10 employees work.

Query :

```
SELECT D.DEPT_NAME
FROM DEPARTMENT D JOIN EMPLOYEE E ON D.DEPT_CODE=E.DEPT_CODE
GROUP BY D.DEPT_CODE,D.DEPT_NAME HAVING COUNT(*) >= 10;
```

```
mysql> SELECT
-> D.DEPT_NAME
-> FROM DEPARTMENT D
-> JOIN EMPLOYEE E
-> ON D.DEPT_CODE=E.DEPT_CODE
-> GROUP BY D.DEPT_CODE,D.DEPT_NAME
-> HAVING COUNT(*) >= 10;
```

```
+-----+
| DEPT_NAME |
+-----+
| Personnel |
| Production|
| Purchase  |
| Finance   |
| Research  |
+-----+
```

5 rows in set (0.00 sec)

Question 31: Find the department code in which employee with highest Basic works.

Query :

```
SELECT DEPT_CODE FROM EMPLOYEE WHERE BASIC = (
    SELECT MAX(BASIC)
    FROM EMPLOYEE
);
```

```
mysql> SELECT DEPT_CODE
-> FROM EMPLOYEE
-> WHERE BASIC =
-> (
-> SELECT MAX(BASIC)
-> FROM EMPLOYEE
-> );
```

```
+-----+
| DEPT_CODE |
+-----+
| DP001     |
+-----+
```

1 row in set (0.00 sec)

Question 32: Find the Designation description of the employee with highest basic.

Query :

```
SELECT D.DESIG_DESC
FROM EMPLOYEE E
JOIN DESIGNATION D
      ON E.DESIG_CODE=D.DESIG_CODE
WHERE E.BASIC =
```

```
(
  SELECT MAX(BASIC)
  FROM EMPLOYEE
);
```

```
mysql> SELECT D.DESIG_DESC
-> FROM EMPLOYEE E
-> JOIN DESIGNATION D
->      ON E.DESIG_CODE=D.DESIG_CODE
-> WHERE E.BASIC =
-> (
->      SELECT MAX(BASIC)
->      FROM EMPLOYEE
-> );
```

```
+-----+
| DESIG_DESC |
```

```
+-----+
| Executive |
```

```
+-----+
1 row in set (0.00 sec)
```

Question 33: Find the no. of managers in each department.

Query :

```
SELECT
      E.DEPT_CODE,
      COUNT(*) AS TOTAL MANAGERS
FROM EMPLOYEE E
JOIN DESIGNATION D
      ON E.DESIG_CODE=D.DESIG_CODE
WHERE D.DESIG_DESC='Manager'
GROUP BY E.DEPT_CODE;
```

```
mysql> SELECT
->      E.DEPT_CODE,
->      COUNT(*) AS TOTAL MANAGERS
-> FROM EMPLOYEE E
-> JOIN DESIGNATION D
->      ON E.DESIG_CODE=D.DESIG_CODE
-> WHERE D.DESIG_DESC='Manager'
-> GROUP BY E.DEPT_CODE;
```

```
+-----+-----+
| DEPT_CODE | TOTAL MANAGERS |
```

```
+-----+-----+
| DP001    | 34              |
| DP002    | 38              |
| DP003    | 39              |
| DP005    | 30              |
| DP004    | 40              |
```

```
+-----+-----+
5 rows in set (0.00 sec)
```

Question 34: Find the maximum basic from EMP table without using MAX().

Query :

```
SELECT BASIC AS MAX_BASIC
FROM EMPLOYEE
ORDER BY BASIC DESC
LIMIT 1;
```

```
mysql> SELECT BASIC AS MAX_BASIC
-> FROM EMPLOYEE
-> ORDER BY BASIC DESC
-> LIMIT 1;
```

```
+-----+
| MAX_BASIC |
+-----+
| 89977.00 |
+-----+
```

1 row in set (0.00 sec)

Question 35: Find the minimum basic from EMP table without using MIN().

Query :

```
SELECT BASIC AS MIN_BASIC
FROM EMPLOYEE
ORDER BY BASIC ASC
LIMIT 1;
```

```
mysql> SELECT BASIC AS MIN_BASIC
-> FROM EMPLOYEE
-> ORDER BY BASIC ASC
-> LIMIT 1;
```

```
+-----+
| MIN_BASIC |
+-----+
| 18000.00 |
+-----+
```

1 row in set (0.00 sec)

Question 36: Find the name of the department with highest total basic. Do the same for highest average basic and maximum no. of employee.

Query :

```
SELECT D.DEPT_NAME
FROM DEPARTMENT D JOIN EMPLOYEE E ON D.DEPT_CODE=E.DEPT_CODE
GROUP BY D.DEPT_CODE,D.DEPT_NAME
ORDER BY SUM(E.BASIC) DESC LIMIT 1;
```

```
SELECT D.DEPT_NAME
FROM DEPARTMENT D JOIN EMPLOYEE E ON D.DEPT_CODE=E.DEPT_CODE
GROUP BY D.DEPT_CODE,D.DEPT_NAME
ORDER BY AVG(E.BASIC) DESC LIMIT 1;
```

```
SELECT D.DEPT_NAME
FROM DEPARTMENT D JOIN EMPLOYEE E ON D.DEPT_CODE=E.DEPT_CODE
GROUP BY D.DEPT_CODE,D.DEPT_NAME
ORDER BY COUNT(*) DESC LIMIT 1;
```

```
mysql> SELECT D.DEPT_NAME
-> FROM DEPARTMENT D
-> JOIN EMPLOYEE E
-> ON D.DEPT_CODE=E.DEPT_CODE
-> GROUP BY D.DEPT_CODE,D.DEPT_NAME
-> ORDER BY SUM(E.BASIC) DESC
-> LIMIT 1;
```

DEPT_NAME
Finance

1 row in set (0.00 sec)

```
mysql> SELECT D.DEPT_NAME
-> FROM DEPARTMENT D
-> JOIN EMPLOYEE E
-> ON D.DEPT_CODE=E.DEPT_CODE
-> GROUP BY D.DEPT_CODE,D.DEPT_NAME
-> ORDER BY AVG(E.BASIC) DESC
-> LIMIT 1;
```

DEPT_NAME
Research

1 row in set (0.00 sec)

```
mysql> SELECT D.DEPT_NAME
-> FROM DEPARTMENT D
-> JOIN EMPLOYEE E
-> ON D.DEPT_CODE=E.DEPT_CODE
-> GROUP BY D.DEPT_CODE,D.DEPT_NAME
-> ORDER BY COUNT(*) DESC
-> LIMIT 1;
```

DEPT_NAME
Finance

1 row in set (0.00 sec)

Question 37: Insert same rows into EMP table with designation code not existing in DESIGNATION table.

Query :

INSERT INTO EMPLOYEE

VALUES

('E100','Test User','DP001','D999','M',
'ABC Road','Kolkata','WB','700001',
30000,'2024-01-01');

```
mysql> INSERT INTO EMPLOYEE
-> VALUES
-> ('E100','Test User','DP001','D999','M',
-> 'ABC Road','Kolkata','WB','700001',
-> 30000,'2024-01-01');
```

ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`assignments`.`employee`, CONSTRAINT `employee\_ibfk\_2` FOREIGN KEY (`DESIG\_CODE`) REFERENCES `designation` (`DESIG\_CODE`))

After dropping the foreign key :

```
mysql> INSERT INTO EMPLOYEE
-> VALUES
-> ('E100','Test User','DP001','D999','M',
-> 'ABC Road','Kolkata','WB','700001',
-> 30000,'2024-01-01');
```

Query OK, 1 row affected (0.01 sec)

Question 38: Delete the rows from EMP table with invalid DESIG_CODE.

Query :

```
DELETE FROM EMPLOYEE WHERE DESIG_CODE NOT IN (
    SELECT DESIG_CODE
    FROM DESIGNATION
);
```

```
mysql> DELETE FROM EMPLOYEE
-> WHERE DESIG_CODE NOT IN
-> (
->     SELECT DESIG_CODE
->     FROM DESIGNATION
-> );
Query OK, 0 rows affected (0.00 sec)
```

Question 39: Find the name of the female employees with basic greater than the average basic of their respective department.

Query :

```
SELECT E1.EMP_NAME FROM EMPLOYEE E1 WHERE E1.SEX='F' AND E1.BASIC > (
    SELECT AVG(E2.BASIC)
    FROM EMPLOYEE E2
    WHERE E2.DEPT_CODE=E1.DEPT_CODE
);
```

```
mysql> SELECT
->     E1.EMP_NAME
-> FROM EMPLOYEE E1
-> WHERE E1.SEX='F'
-> AND E1.BASIC >
-> (
->     SELECT AVG(E2.BASIC)
->     FROM EMPLOYEE E2
->     WHERE E2.DEPT_CODE=E1.DEPT_CODE
-> );
```

EMP_NAME
Rohit Singh
Anirban Pillai
Akash Nandi
Pallavi Kapoor
Ritika Patel
Debjani Agarwal
Santanu Bera
Payel Debnath
Prasenjit Barman

Question 40: Find the number of female managers.

Query :

```
SELECT COUNT(*) AS FEMALE MANAGERS FROM EMPLOYEE E JOIN DESIGNATION D ON
E.DESIG_CODE=D.DESIG_CODE WHERE E.SEX='F' AND D.DESIG_DESC='Manager';
```

```
mysql> SELECT COUNT(*) AS FEMALE MANAGERS
-> FROM EMPLOYEE E
-> JOIN DESIGNATION D
-> ON E.DESIG_CODE=D.DESIG_CODE
-> WHERE E.SEX='F'
-> AND D.DESIG_DESC='Manager';
```

FEMALE MANAGERS
91

1 row in set (0.00 sec)